Calibration Program Method Guideline

Prerequisites

Before you begin, ensure the following are prepared and ready:

- A circle grid calibration board (BDB-6) securely mounted on the cobot TCP.
- A working camera connected to the system and positioned to view the TCP area.
- An ARC Cobot connected and controllable via network.
- The Python environment and all required dependencies installed.

Step-by-Step Guide:

This section explains how to configure and run the collect_calibration_data.py script to capture images for calibration purposes.

1. Collecting Calibration Data

This guide explains how to configure and run the collect_calibration_data.py script to capture images for camera calibration.

Step 1: Open the Script

Locate and open the file: collect_calibration_data.py

Step 2: Update Configuration Settings

Inside the script, modify the following parameters:

a. File Name for Saved Images

   Set the filename or folder where captured images will be stored.

b. Camera IP Address

   Update the IP to match your connected camera device.

c. Robot and PC IP Address

Set the pc and robot's IP address for motion control.

Examples:

a. File name : mechmind_logMleft_25jun
b. Camera IP address : 192.168.100.205
c. Robot IP address : 192.168.100.210
d. PC IP address : 192.168.100.200

```
calibration_folder_name = os.path.join("data", "mechmind_logMleft_25jun")

robot_port = 30001
robot_port_pose = 30001
pc_port = 30005
camera_ip = "192.168.100.205"
robot_ip = '192.168.100.210'
pc_ip = '192.168.100.200'
```

Step 3: Define Starting Pose

Physically move the robot so that:

- The calibration board is centered in the camera's field of view.
- The Z-axis is perpendicular (normal) to the board.

Update the starting pose (XYZ and orientation) in the script:

This point will be used as the initial pose from which automated path movement begins.

Example :

Robot starting position : 100, -600, -100

```
calibration_folder_name = os.path.join("data", "mechmind_logMleft_25jun")

robot_port = 30001
robot_port_pose = 30001
pc_port = 30005
camera_ip = "192.168.100.205"
robot_ip = '192.168.100.210'
pc_ip = '192.168.100.200'
x, y, z = 100, -600, -100
```

Step 4: Run the Script

Once all settings are configured, run the script

Step 5: Verify Captured Images

The script will automatically capture and save images as the robot moves.

Check the saved files in the folder defined earlier (e.g., mechmind_logMleft_25jun) to ensure:

- All circles on the calibration target are clearly visible.
- The depth map of the circle grid is present and correctly rendered.

Step 6: Adjust as Needed

The image capture path is generated automatically.

If the board moves out of view or circles are not properly visible:

- Press the E-stop to halt the robot immediately.
- Adjust the robot's starting position or orientation.
- Update the corresponding starting pose in the script.

2. Calibration grid selecting

This guide explains how to use calibration_grid_select_blob.py to label the calibration circles in the captured images for camera calibration.

Step 1: Open the Script

Locate and open the file: calibration_grid_select_blob.py

Step 2: Update the Image Folder Path

In the script, modify the path to point to the folder containing the saved calibration images.

Example :

Image folder : mechmind_logMright_25jun

```
5
6   calibration_folder_path = os.path.join("data","mechmind_logMright_25jun")
7
```

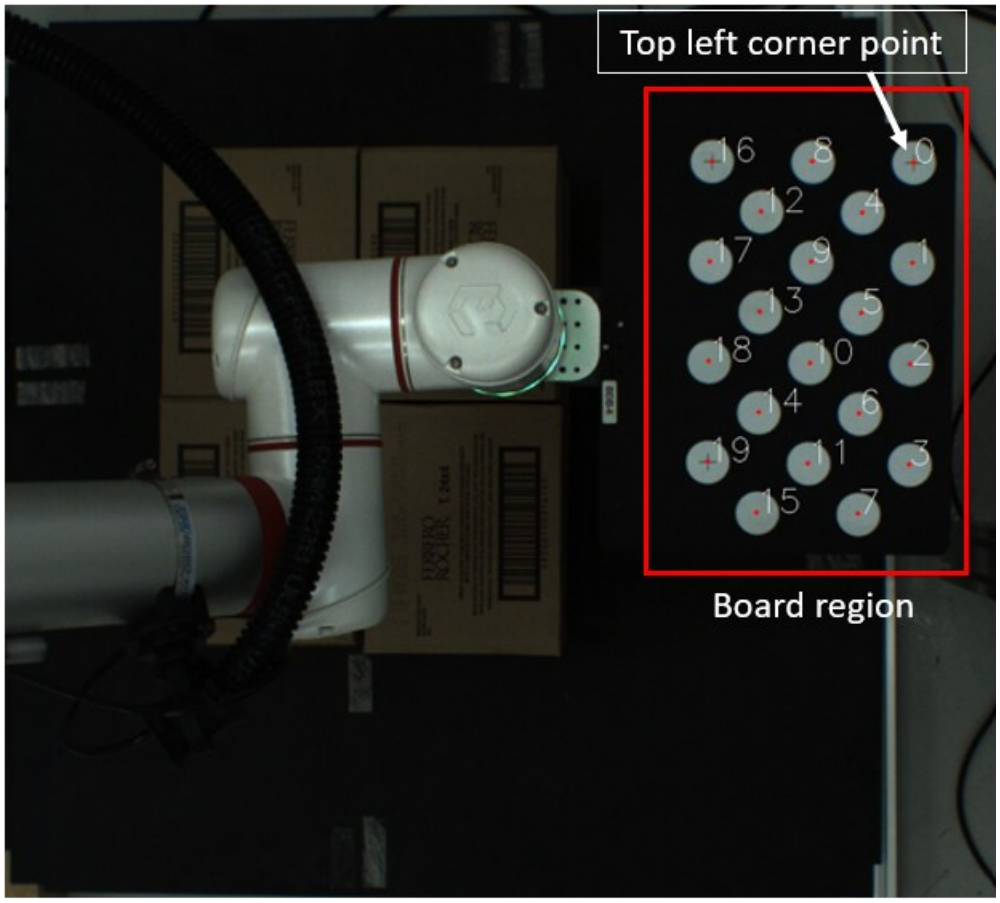
Step 3: Run the Script

Once all settings are configured, run the script

Step 4: Select Calibration Circles

For each image that appears:

a. The script will automatically detect and highlight all detected circles.
b. To select the calibration grid:
   - Click and drag with the left mouse button to draw a rectangle around the grid area.
   - Ensure that only the calibration circles are included in the region.
c. check the sequence numbers — they should start from the top-left circle and follow a row-wise order (left to right, then top to bottom).



⚠️ **Important**:

If the region contains too few or extra circles, the script will prevent you from proceeding to the next step. In that case, reselect the region correctly.

Step 5: Select Top-Left Corner Circle

After selecting a valid region:

- Click on the top-left circle of the calibration board as shown in the previous image
- The top-left circle refers to the one located on the side with three circles and is the farthest from the TCP.
- This defines the orientation of the grid.

If you selected the wrong point:

- Right-click to restart the labeling for that image.

Step 6: Proceed or Skip

Once the region and corner are correctly selected:

- Press the spacebar to proceed to the next image.

If an image is incomplete (e.g., not all circles are visible):

- You may skip the image by pressing the spacebar without labeling.

Calibration grid selecting

Step 1: Open the Script

Locate and open the file: run_computation_scripts

Step 2: Update the Image Folder Path

In the script, modify the path to point to the folder containing the saved calibration images.

Example :

Image folder : mechmind_logMright_25jun

Copy and paste a camera.npz file based on the type of camera used from manufacturer_camera_parameters into the data file and rename it as camera_parameters.npz

Step 3: Run the Script

Once all settings are configured, run the script

After execution, three files will be generated in the output folder:

- optimized.npz
- board_alignment.npz
- extrinsic.npz

Take the optimized.npz file, rename it appropriately, and save it into the camera_parameters folder. This completes the calibration process.